Customer Churn Analysis — Executive Summary Report

Telecommunications Industry | IBM Telco Dataset | Predictive Analytics

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1. BUSINESS PROBLEM

Customer churn, when a customer discontinues a subscription or service, represents one of the most significant revenue challenges in the telecommunications industry. Acquiring a new customer costs **5 to 7 times more** than retaining an existing one. With a dataset of **7,043 Telco customers**, this analysis was conducted to identify the key drivers of churn, build predictive machine learning models, and provide data-driven recommendations to reduce churn and protect revenue.

2. KEY PERFORMANCE INDICATORS

TOTAL CUSTOMERS	CHURN RATE	MONTHLY REVENUE AT RISK	HIGH RISK CUSTOMERS
7,043	**26.5%**	**$139K**	**1,997**
IBM Telco Dataset	1 in 4 customers churns	Lost per month to churn	Churn probability > 60%

3. KEY FINDINGS

Contract Type

Month-to-month customers churn at **42.7%** — 14x higher than two-year contract customers (3%). Contract type is the single strongest retention lever in the dataset.

Customer Tenure

47% of churners leave within the first 6 months. Churn rate drops from 53% at 0-6 months to just 7% at 61-72 months. The first 6 months is the most critical retention window.

Monthly Charges

Churned customers pay on average **$74.44/month** vs $61.27 for retained customers. High-paying customers feel they are not receiving sufficient value.

Internet Service

Fiber optic customers churn at **41.9%** — more than double the DSL rate (19%). Suggests unmet expectations around speed, reliability or pricing for Fiber customers.

Payment Method

Electronic check users churn at **45%** — the highest of all payment methods — compared to only 15% for automatic credit card payments. Manual payment correlates strongly with lower engagement.

Add-on Services

Customers without Tech Support or Online Security churn at nearly **double the rate** of those with add-ons (41% vs 21%). Add-ons increase perceived value and stickiness.

4. MACHINE LEARNING MODEL RESULTS

Three machine learning models were trained and evaluated using Python (Scikit-learn, XGBoost). Models were compared using **AUC-ROC** as the primary metric — measuring the ability to correctly rank a churner above a non-churner across all probability thresholds.

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	74.0%	50.7%	78.6%	61.6%	0.841
Random Forest (Best Model)	**76.9%**	**54.9%**	**74.1%**	**63.0%**	**0.842**
XGBoost	76.1%	53.6%	74.1%	62.2%	0.835

Best Model: Random Forest — *selected based on highest AUC-ROC (0.842). Top predictors: tenure, monthly charges, contract type and internet service.*

5. BUSINESS RECOMMENDATIONS

IMMEDIATE — High Risk Customers (1,997)

- Contact all 1,997 High Risk customers (churn probability > 60%) within 48 hours via personal outreach.
- Offer targeted retention incentive — 15-20% discount or service upgrade — framed as a loyalty reward.
- Prioritise customers with tenure 1-6 months, Fiber Optic internet and month-to-month contracts.

SHORT TERM — Contract Strategy

- Launch a contract upgrade campaign offering month-to-month customers a discounted annual plan.
- Implement a lock-in incentive at month 3 and month 6 — the highest churn windows — with a free month or service bundle.
- Set up automated alerts for any month-to-month customer who has not logged in for 14+ days.

LONG TERM — Product and Service Improvements

- Investigate Fiber Optic dissatisfaction — conduct NPS surveys to identify speed and reliability gaps.
- Bundle Tec Support and Online Security with all Fiber Optic plans to increase perceived value and reduce churn.
- Introduce auto-pay incentives (small discount) to shift electronic check users to automatic payment methods.

Tools and Technologies	Python (Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn) | Power BI Desktop | Google Colab | IBM Telco Dataset